

Nederman Ducting

Field Labeling Guide

How to mark, key and sequence duct before it leaves the shop

Per Karl's direction, every piece of Nederman duct gets labeled before it ships. This is the standard: what goes on the piece, how the runs are keyed, and how the numbers are sequenced so a crew can lay the whole system out on the floor and build it in order without a print in their hand.

1. The five rules

Everything below comes out of these. If you remember nothing else, remember these five.

R1	Write the diameter on every piece. No exceptions — straights, elbows, reducers, laterals, blast gates, hoods. A piece with no diameter on it is an unlabeled piece.
R2	Sequence small to large. Numbering starts at the smallest piece farthest from the fan and counts up toward the trunk. The crew installs in that same order.
R3	Key the ducts. Every run gets a letter off the drawing. Main trunk is A. Laterals and Y-branches get their own letter, not a share of the main's.
R4	Above 25 goes to flange. Anything larger than 25 is flanged construction, not quick-clamp. Mark the size on the flange face so it stays readable once the piece is rigged and the body is against the ceiling.
R5	Mark both ends. One end is always the end you can't see once it's up.

Confirm the units before the first crate ships

The sketch reads 32 / 24 / 16 and the rule reads “above 25.” That is consistent with the Nederman quick-connect break — clamp-together up to 250 mm, flanged at 280 mm and above — and it is equally consistent with inches. This guide writes the bare number the way the sketch does, so it is correct either way. Pick one unit for the whole job, state it once on the drawing, and never mix the two on a label.

2. The label

Every piece carries the same four fields, in the same order:

SYSTEM	RUN	SEQ	DIA
DC1	A	03	Ø32
which collector / system it belongs to	letter off the drawing	build order, small → large	outside diameter, always with Ø

Written out on the piece:

DC1-A-03-Ø32

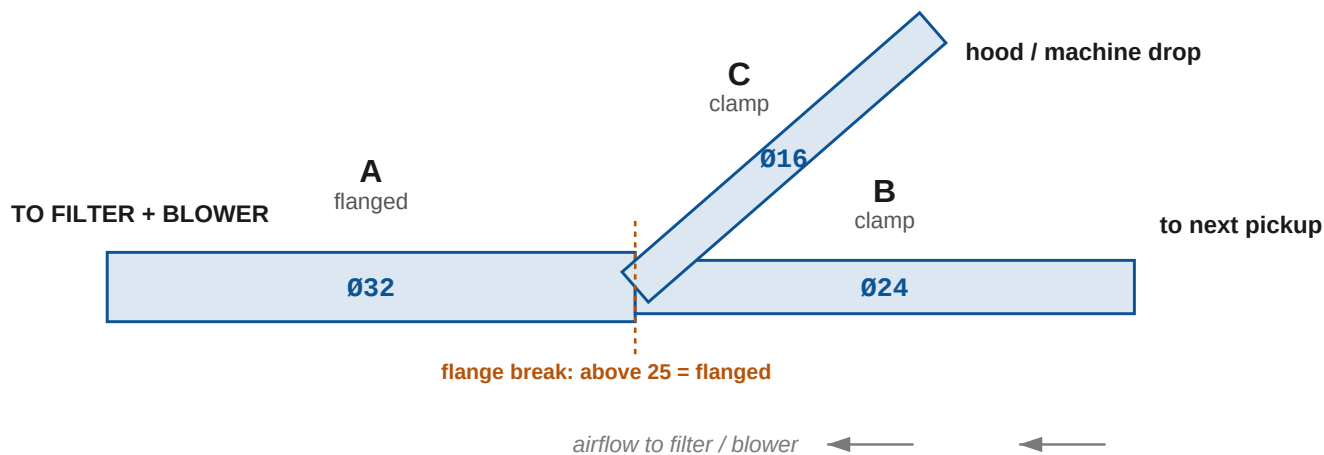
Add length in inches after the diameter when the piece is a cut straight and two lengths on the job could be confused — **DC1-A-03-Ø32-96**. Fittings don't need a length; they need their type instead (see §5).

Assigning the letters

- **A is the trunk at the collector** — the biggest pipe, closest to the filter and blower. Letters walk outward from there: A, then B, then C.
- **A lateral or Y gets the next free letter**, not a sub-number of the main. A 16 branch dropping off the 24 main is run C, not B-2. When the crew is hunting a piece on a crowded floor, a letter is faster to find than a suffix.
- **The letter never changes size mid-run.** If the diameter steps, the run ends and a new letter starts at the reducer. That is what makes the label self-checking: two pieces with the same letter and different diameters means somebody mislabeled one of them.

3. Worked example — Karl's sketch

This is the branch off the field sketch: a 32 trunk running to the filter and blower, a 24 main continuing out to the next pickup, and a 16 lateral dropping to a machine hood. Three sizes, three letters.



Run	Dia	What it is	Connection	Label reads
A	32	Trunk — junction back to filter / blower	Flanged (>25)	DC1-A-03-Ø32
B	24	Main — junction out to next pickup	Quick clamp	DC1-B-02-Ø24
C	16	Lateral / Y — drop to machine hood	Quick clamp	DC1-C-01-Ø16

Read the sequence numbers and the build order falls out of them: hang C first (01, the 16 at the hood), then B (02, the 24 main), then A (03, the 32 into the collector). Small to large, exactly as Karl called it — the crew works from the machines back to the fan, and the heavy flanged pipe goes up last, when the lift already has clear floor under it.

The Y itself gets labeled too, on the body, with all three legs called out: **DC1-Y-Ø32xØ24xØ16**. Write the legs in the order trunk × main × branch so the fitter knows which port faces the fan before they wrestle it onto the strut.

4. Where and how to mark

Situation	Where the mark goes
Straight duct, 25 and under (clamp)	Both ends, on the outside of the pipe, roughly 6" back from each end so the clamp band doesn't cover it. Repeat the full label once at mid-length for anything over 8 feet.
Straight duct, above 25 (flanged)	On the flange face , per R4 — both flanges — plus once on the body. The flange stays visible from the floor after the piece is rigged; the body may not.
Elbows, reducers, laterals, Y	On the body, on the outside of the sweep where it will face down or out once installed. Never on the inside of a bend.
Blast gates, dampers	On the frame, next to the handle — not on the blade or the moving plate.
Hoods and machine drops	On the collar, plus the machine name or number the hood serves. This is the one label the plant's own maintenance crew will read for the next ten years.

Marking method

- **Silver or white paint marker on galvanized and painted duct; black on stainless or bare bright.**
Permanent marker fades off galv in a few weeks of shop light and disappears completely on an outdoor lay-down yard.
- **Print, don't script.** Caps, 2" tall minimum on anything 16 and up. It has to be readable from the floor when the piece is on the ceiling.
- **Back the paint marker up with a tag on anything that ships loose** — wire-on weatherproof tag with the same label written on it. Paint gets scuffed off in transit; a tag survives.
- **Never label over a seam, weld, or the sealing face of a gasket.** Paint on a flange gasket face is a leak path.

5. Fittings — what else goes on the label

Straights are identified by diameter and length. Fittings are identified by diameter and *type*, because two 24 elbows are not interchangeable if one is a 45 and one is a 90.

Fitting	Add to the label	Example
Elbow	Angle and centerline radius	DC1-B-04-024-90°-R1.5D
Reducer	Both diameters, large first	DC1-A-05-RED-032x024
Lateral / Y	All legs: trunk × main × branch	DC1-Y-032x024x016
Blast gate	Diameter and the machine it isolates	DC1-C-BG-016-SANDER 2
Hood / drop	Collar diameter and machine	DC1-C-HD-016-SANDER 2
Flex hose	Diameter and cut length	DC1-C-FLX-016-36

6. Crates, bundles and the pack list

- **Crate by run, not by size.** One crate holds run C complete — duct, elbows, gate, hood, clamps, gaskets. The crew opens one box and hangs one branch. Sorting a crate of mixed 16s across four runs is where a day gets lost.
- **Stencil the crate with the system and the runs inside — DC1 / RUN C / PCS 01-07** — on two adjacent sides, so it reads however the forklift sets it down.
- **Pack list goes inside the crate and a copy in the shipping sleeve**, listing every label in the crate. That list is the receiving check and the punch list for anything short-shipped.
- **Load in reverse install order.** Highest sequence numbers go in the truck first so the smallest, first-installed pieces come off the truck first.

7. Before it ships — QC check

☐	Every piece has a diameter on it, both ends.
☐	Every piece above 25 is flanged, and the size is marked on the flange face.
☐	Run letters match the drawing exactly — no run letter reused, no size change inside a letter.
☐	Sequence numbers run small to large with no gaps and no duplicates.
☐	Fittings carry their type: angle, both reducer sizes, all three Y legs.
☐	Every hood and blast gate names its machine.
☐	Marks are paint, not permanent marker, and legible from twenty feet.
☐	Loose pieces carry a wire tag matching the painted label.
☐	Crates stencilled on two sides; pack list inside and in the sleeve.
☐	A marked-up copy of the drawing ships with the load, showing the same letters and numbers as the pieces.

8. The mistakes that cost a day

Mistake	What it costs
Labeling on the lift instead of in the shop	Two people and a lift tied up doing paperwork at height. Label on the bench where it takes seconds.
Marking one end only	The unmarked end goes up, and the next fitter has to guess or take a tape to it in the air.
Numbering large to small	Crew starts at the fan and works out. Every joint downstream then has to carry the weight of everything not yet hung.
Reusing a run letter after a size change	Two different pieces answer to the same label. Guaranteed wrong piece on the lift at least once.

Mistake	What it costs
Mixing units on one job	A “25” that means millimeters in the shop and inches in the field is a scrapped fitting.
Trusting the label and skipping the drawing	Labels tell you what a piece is, not where it goes. The marked-up drawing ships with the crate for a reason.

9. Bench card

Print this page, cut on the line, pin it over the layout bench.

EVERY PIECE, EVERY TIME

DC1-A-03-Ø32

system – run – seq – diameter

Both ends. Paint marker, caps, 2” tall.

Above 25 → flanged. Size goes on the flange face.

Fittings add their type: 90°, RED-Ø32xØ24, Y-Ø32xØ24xØ16.

Gates and hoods add the machine name.

THE FIVE RULES

1. Write the diameter on every piece.

2. Sequence small to large — 01 is the far pickup, the last number is the trunk at the fan.

3. Key every run with a letter. A = trunk. A lateral or Y gets its own letter.

4. Above 25 goes to flange — mark the flange.

5. Mark both ends.

One letter = one diameter. If the size steps, the run letter steps with it.

Rev A — drafted from Karl's labeling direction and the field sketch (32 trunk / 24 main / 16 lateral). Confirm the unit convention and the exact flange break with Karl before the first crate ships, then this becomes the shop standard for all Nederman duct.